Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Next-generation Tumor Endothelial Cell–Targeted Therapeutics in Refractory Solid Tumors			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (V), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio/Medical		Bio/Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Therapeutic Antibodies	Industrial Biotechnology	Bio-engineering Technologies
Concept of Proposed Technology	Current cancer therapies rely on disparate targeting of epithelial cell growth, angiogenesis, or T cell activation. This leads to failure to adequately address immune exclusion, abnormal vasculature, and microenvironment-driven therapeutic resistance. These limitations are evidenced by low response rates (e.g., <10-30%) in most therapeutics that target vascular endothelial growth factor (VEGF) pathways and/or immune checkpoints. Furthermore, technologies that are not specific to the tumor microenvironment (TME) lead to adverse effects. New strategies to reprogram the vascular-immune axis is critically warranted. The proposed technology should be designed to selectively modulate **tumor endothelial cell–driven mechanisms, distinct from traditional VEGF/VEGF-receptor targets,** that contribute to pathological angiogenesis, tumor hypoxia, immune suppression, and physical barriers to immune cell infiltration. Platforms that integrate endothelial and immune targets, in particular, by way of **developing bispecific antibodies**, are expected to offer tumor-specific normalization of tumor vasculature, restoration of immune cell access, and enhanced antitumor immune responses. This dual-action			

	strategy has the potential to improve responsiveness to existing immunotherapies and establish a differentiated therapeutic paradigm for refractory solid tumors. The proposed technology will be broadly applicable across refractory solid tumors and may also be extendable to other angiogenesis-related diseases beyond oncology.
Rationale for Joint R&D Project	Refractory solid tumors remain a major unmet medical need, as many patients exhibit intrinsic or acquired resistance to immune checkpoint inhibitors and anti-angiogenic therapies. Existing VEGF- and PD-1–based approaches often fail to sufficiently normalize tumor vasculature or reverse immune exclusion, resulting in limited and unsustained clinical benefit. A joint R&D framework involving research institutions and industrial partners is essential to: • Translate advanced insights in vascular biology and tumor microenvironment regulation into differentiated therapeutic candidates • Share technical and developmental risks inherent in early-stage biologics innovation • Accelerate global competitiveness through validation of first-in-class or highly differentiated mechanisms of action • Strengthen Korea's advanced bio ecosystem through international collaboration and cross-border R&D integration
Key Words	1. Tumor Endothelial Cells (TEC) 2. Tumor Microenvironment (TME) 3. Vascular Normalization 4. Antibody Therapeutics 5. Refractory Solid Tumor

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3~4)	**End of the Project**	Level (5~6)
R&D Objectives	• Validate a tumor endothelial cell–specific therapeutic target with mechanistic relevance to pathological angiogenesis, immune exclusion, and solid tumor progression • Develop bispecific antibody candidates incorporating both tumor endothelial target and immune checkpoint, e.g., PD-1/PD-L1, designed to selectively modulate tumor vasculature and immune responses • Demonstrate robust in vitro and in vivo efficacy, including vascular normalization and immune microenvironment reprogramming in models of immunotherapy resistance • Establish a scalable and translational development pathway supporting advancement toward clinical-stage biologics			
R&D Target Specifications	**Target Characteristics** • Preferential expression in tumor endothelial cells with minimal expression in normal vasculature • Functional involvement in pathological angiogenesis, immune exclusion, and tumor progression • Translational relevance between preclinical models and human disease			

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	Bispecific Antibody Design Requirements • Incorporation of immune checkpoint targeting, e.g., PD-1 or PD-L1, as one binding arm and a tumor endothelial cell–specific target as the second arm • Molecular format supporting stability, manufacturability, and standard antibody engineering processes **Binding and Selectivity** • Binding affinity: KD ≤ 10 nM for each binding arm • ≥10-fold preferential binding to tumor-associated endothelial cells versus normal endothelial cells • Minimal disruption of physiological vascular homeostasis **Functional Outcomes** • Evidence of tumor-specific vascular normalization • Reduction of tumor hypoxia and pathological angiogenesis • Increased immune cell infiltration and activation within the tumor microenvironment **In Vitro and In Vivo Efficacy** • Demonstrated endothelial and immune-related functional activity in vitro • Statistically significant in vivo efficacy in validated solid tumor models, including activity in models resistant to checkpoint immunotherapy **Safety and Developability** • No significant adverse impact on cardiovascular or systemic vascular function in preliminary assessments • Compatibility with scalable manufacturing and future GMP production
R&D Plan	**Overall Approach** The project follows a **stage-gated, milestone-driven development plan** over three years, progressing from target validation to preclinical lead selection and translational readiness. --- *Phase 1: Target Validation and Bispecific Antibody Design (Year 1)* • Validation of tumor endothelial cell–specific target expression and function • Design and generation of immune checkpoint (PD-1/PD-L1) bispecific antibody prototypes • In vitro binding, selectivity, and functional characterization **Key Milestones:** • Confirmed tumor-specific endothelial target • Bispecific antibody prototypes generated • In vitro proof-of-mechanism demonstrated *Phase 2: Lead Optimization and Preclinical Efficacy (Year 2)* • Optimization of potency, selectivity, stability, and developability • Efficacy in model systems that recapitulate the TME • In vivo efficacy evaluation in relevant, immunocompetent mouse models of solid tumors

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- Assessment of vascular normalization and immune microenvironment modulation

Key Milestones:

- Optimized bispecific antibody lead(s) identified
- Statistically significant in vivo anti-tumor efficacy demonstrated

Phase 3: Translational Development and Preclinical Readiness (Year 3)

- Exploratory pharmacokinetic and preliminary safety evaluation
- Assessment of manufacturing feasibility and formulation stability
- Development of translational strategies and biomarker concepts

Key Milestones:

- Selection of a preclinical lead candidate
- Completion of a development package suitable for IND-enabling advancement

Summary Timeline

Year	Major Focus	Key Deliverables
Year 1	Target validation & bispecific design	Validated target, bispecific prototypes, in vitro PoC
Year 2	Lead optimization & in vivo efficacy	Optimized lead(s), in vivo efficacy & MOA data
Year 3	Translation & readiness	Preclinical lead, PK/safety data, IND-readiness strat

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Characterize the expression of a given target for reprogramming the tumor vasculature in lethal, immunotherapy-resistant cancer contexts, e.g., pancreatic cancer • Conduct *in vitro* and *in vivo* proof-of-concept studies related to vascular-immune microenvironment targeting • High-parameter spatial determination of tumor microenvironment changes upon target perturbation to nominate translational biomarkers and pharmacodynamic correlatives • Explore patient stratification strategies by characterizing the target expression within a given TME context in a pilot cohort of clinical specimens • Provide scientific guidance, data interpretation, and participation in joint decision-making
Expected R&R of Korean SME Partner	• Lead engineering and optimization of endothelial-targeting plus immune checkpoint (e.g., PD-1/PD-L1) bispecific antibody candidates • Conduct preclinical efficacy, pharmacology, and developability assessments • Establish manufacturing feasibility and scale-up strategies • Manage project coordination, documentation, and milestone tracking • Support preparation for regulatory and clinical development pathways

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological and Scientific Outcomes • Establishment of a differentiated immune checkpoint bispecific antibody platform targeting tumor endothelial cells • Validation of a novel therapeutic paradigm integrating vascular normalization and immune checkpoint modulation • Generation of robust preclinical proof-of-concept data supporting clinical advancement
	Commercialization and Industrial Impact • Creation of high-value biologic assets suitable for global licensing or co-development • Strengthening of Korean SME capabilities in advanced biologics and immuno-oncology • Expansion potential into additional angiogenesis-related disease areas
	Societal and Healthcare Impact • Advancement of novel therapeutic options for patients with refractory solid tumors • Potential improvement in long-term clinical outcomes through tumor microenvironment reprogramming • Contribution to national strategic objectives in advanced bio convergence and global therapeutic innovation

6. References

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(3) Schaaf, M.B.E., Garg, A.D., & Agostinis, P. "*Defining the role of the tumor vasculature in antitumor immunity and immunotherapy",* Cell Death & Disease, 9, 115 (2018).

(4) Allen, E. et al. "*Combined antiangiogenic and anti–PD-L1 therapy stimulates tumor immunity through HEV formation",* Science Translational Medicine, 9(385), eaak9679 (2017).

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